



Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no.....

Name of the Course: MBA

Semester: IV

Name of the Paper: Data Science Using R

Paper Code: MBA 404 (BA1)

Time: 1.5 hour

Maximum Marks: 50

Note:

- (i) This question paper contains two Sections - A and B
- (ii) Both Sections are compulsory
- (iii) Answer any two sub questions from a, b & c in each main question of Section A. Each sub question carries 5 marks.
- (iv) Section B, consisting of a case study, is compulsory. It is of 20 Marks.

Section A

Q1.

(2X5 = 10 Marks)

- a. Explain the importance and scope of Data Analytics in business decision making with suitable real-life examples. CO1
- b. Develop R expressions and user-defined functions to perform arithmetic or logical operations on datasets. CO3
- c. Write R commands to install packages, load libraries, and execute R scripts for performing data analysis tasks. CO2

Q2.

(2X5 = 10 Marks)

- a. Describe the responsibilities of a Data Scientist and explain how R programming supports these roles. CO1
- b. Explain and demonstrate the use of R data structures (Vectors, Matrix, Arrays, Lists, Data Frames) with suitable examples. CO3
- c. Demonstrate the process of setting up RStudio, customizing the interface, and managing projects for data analysis. CO2

Q3.

(2X5 = 10 Marks)

- a. Explain string manipulation techniques in R and demonstrate their use in data preprocessing tasks. CO4
- b. Discuss data munging, sampling, scraping, and data cleaning techniques in R for handling large datasets with examples. CO4
- c. Compare Descriptive, Diagnostic, Predictive, and Prescriptive Analytics and identify appropriate business scenarios for each. CO1

Section B

Q4. Case Study

(20 Marks)

A mid-sized retail store collects daily sales data including **Product ID, Category, Quantity Sold, Price, Customer Type, and Date**. The management wants to understand sales patterns and improve decision making using **Data Analytics and R programming**.

A junior data analyst is assigned to:

- Set up the **RStudio** environment
- Import raw sales data
- Clean inconsistent entries (missing values, spelling issues in product names)
- Convert variables into proper formats
- Use **R data structures** like vectors and data frames
- Create functions to calculate total revenue



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- Perform basic string operations on product categories
- Conduct data munging and sampling to handle large datasets

The analyst must prepare the dataset for further analysis. As a data analyst answer the following questions

- a) Suggest appropriate R data structures for storing the sales dataset and justify your choice. CO3
- b) Write the logical steps for data cleaning and munging of the sales data using R techniques. CO4